Problem 1: Hospital Patient Health Monitoring

1. Mean and sample standard deviation of systolic blood pressure

BP values: 150, 148, 145, 142, 138, 134, 130, 126, 122, 118.

Mean = 135.3 mmHg. Sample standard deviation = 11.14 mmHg (sample variance = 124.01).

2. Covariance

Variable pair	Sample covariance
Exercise and systolic BP	−168.06
BMI and systolic BP	33.61

3. Correlation coefficients

Variable pair	Pearson r	Interpretation
Exercise and BP	−0.9969	Very strong negative relationship
BMI and BP	0.9969	Very strong positive relationship

4. Stronger association: Exercise and BMI have equal absolute correlations with BP in this dataset (about 0.997), but in opposite directions. Exercise is negatively associated with BP, while BMI is positively associated. These values describe association, not proof of causation.

5. Healthcare interventions: encourage safe, regular physical activity; assess BMI and other cardiovascular risk factors; provide nutrition counselling and routine BP monitoring; individualize plans with clinicians.

Problem 2: E-Commerce Customer Behaviour Analysis

1. Purchase amount summary

Mean = ₹4,970. Sample variance = 8,793,444.44 rupees². Sample standard deviation = ₹2,965.37.

2–3. Covariance and correlation

Variable pair	Sample covariance	Pearson r
Website visits and purchase amount	22,568.89	0.9914
Time spent and purchase amount	47,266.67	0.9949

4. Time spent has the slightly stronger linear association with purchase amount (r = 0.9949 versus 0.9914). The difference is small and this dataset alone cannot establish predictive performance or causation.

5. Marketing strategies: improve useful product information and checkout flow; personalize recommendations; test engagement campaigns; use A/B tests and validate predictors on new customers before scaling.

Problem 3: Employee Productivity and Performance

1. Performance rating summary: mean = 80.40; sample variance = 167.82; sample standard deviation = 12.95.

Variable pair	Sample covariance	Pearson r
Training hours and rating	97.87	0.9957
Projects completed and rating	39.11	0.9972

4. Projects completed has the slightly stronger correlation with performance rating (r = 0.9972), compared with training hours (r = 0.9957). The predictors are likely related to each other, and the small dataset cannot establish that either factor causes performance.

5. HR recommendations: provide relevant training, balanced project assignments, mentoring, clear performance criteria, and workload monitoring. Evaluate interventions over time and avoid rewarding project quantity at the expense of quality.

Problem 4: Smart City Traffic Analysis

1. Descriptive statistics (sample variance and sample standard deviation)

Variable	Mean	Sample variance	Sample SD
Traffic volume	18,600	22,155,555.56	4,706.97
Rainfall (mm)	22.50	229.17	15.14
Average speed (km/h)	39.80	97.73	9.89
Accident count	8.00	23.33	4.83

2. Covariance matrix

	Traffic	Rainfall	Speed	Accidents
Traffic	22,155,555.56	32,500.00	−46,722.22	22,611.11
Rainfall	32,500.00	229.17	−148.33	72.22
Speed	−46,722.22	−148.33	97.73	−47.22
Accidents	22,611.11	72.22	−47.22	23.33

3. Correlation matrix

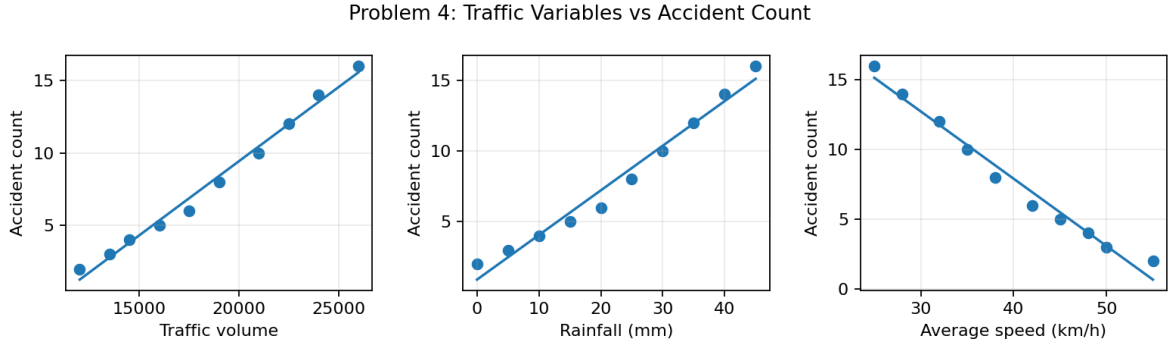
	Traffic	Rainfall	Speed	Accidents
Traffic	1.000	0.456	−0.999	0.994
Rainfall	0.456	1.000	−0.992	0.988
Speed	−0.999	−0.992	1.000	−0.989
Accidents	0.994	0.988	−0.989	1.000

4–5. Accident count is strongly positively associated with traffic volume ($r = 0.994$) and rainfall ($r = 0.988$), and strongly negatively associated with average speed ($r = -0.989$). Traffic volume has the largest absolute correlation with accidents in this dataset.

6. Recommendations: improve junction control and speed management; deploy extra traffic enforcement during heavy rain and peak traffic; maintain drainage and road surfaces; investigate accident hotspots. These are observational data and the variables move together, so correlation alone does not identify a cause.

7. Scatter plots and interpretation

The three scatter plots below show the relationships with accident count. The pattern is nearly linear in this small dataset; collect more days and account for time, road type, visibility and other confounders before making causal claims.



Problem 5: Average Delivery Time

1. Point estimate of population mean = sample mean = 23.6 minutes.

2. Point estimate of population variance = sample variance = 11.40 minutes² (using n−1 in the denominator). Sample SD = 3.38 minutes.

3. Interpretation: the observed orders average 23.6 minutes. The sample standard deviation indicates typical spread around the mean; estimates may vary if another sample of orders is collected.

Problem 6: Customer Spending

1. Estimated average spending = $145.83.

2. Estimated sample standard deviation = $16.07.

3. These are point estimates because each gives one sample-based value for an unknown population parameter (population mean and population SD).

Problem 7: Product Ratings

1. Sample mean = 4.15 stars.

2. Sample variance = 0.5553 stars².

3. Standard error of the mean = $s/\sqrt{n} = 0.7452/\sqrt{20} = 0.1666$ stars.

4. The standard error estimates how much the sample mean would typically vary across repeated samples of 20 ratings. It is not the spread of individual ratings.

Problem 8: Exam Scores

1. Mean = 75.60 points; sample standard deviation = 3.96 points.

2. Standard error = $3.96/\sqrt{25} = 0.792$ points.

3. Standard deviation describes variation among student scores; standard error describes uncertainty in the estimated mean. The SE is smaller because it divides the SD by $\sqrt{n}$.

Problem 9: Manufacturing Weights

Given n = 50, mean = 498 g and sample SD = 12 g. Use a t interval with 49 degrees of freedom.

95% confidence interval = $498 \pm t(0.975,49) \times 12/\sqrt{50}$ = (494.59 g, 501.41 g).

Interpretation: the interval procedure would capture the true population mean in about 95% of repeated samples. Since 500 g lies within this interval, the target is plausible at the 5% level; this does not prove the true mean equals 500 g.

Problem 10: Customer Satisfaction

Branch	Mean	95% CI
A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

The intervals overlap from 4.1 to 4.2. Branch A has the higher sample mean, but overlap of separate 95% confidence intervals is not by itself a formal test of the difference. We cannot conclude confidently that A performs better without a direct comparison test and information about sample sizes and dependence.

Problem 11: Average Battery Life

Hypotheses (two-sided): H_0: $\mu = 10$ hours; H_1: $\mu \neq 10$ hours.

Given $n = 36$, $\bar{x} = 9.5$, $s = 1.8$. Test statistic $t = (9.5-10)/(1.8/\sqrt{36}) = -1.667$, $df = 35$. Two-sided p-value ≈ 0.105.

At $\alpha = 0.05$, fail to reject H_0 because $p > 0.05$. The sample does not provide sufficient evidence that the mean battery life differs from 10 hours; it does not prove the claim is exactly true.

Problem 12: Marketing Campaign

Assume before/after observations are paired for the same comparable periods. Test H_0: mean difference = 0 versus H_1: mean difference > 0.

Mean increase = 8 sales units. Paired differences have SD ≈ 1.195; $t = 18.93$ with $df = 7$. One-sided p-value $\approx 1.43 \times 10^{-7}$ (two-sided $\approx 2.85 \times 10^{-7}$).

Decision: reject H_0 at $\alpha = 0.05$. The after-campaign values are significantly higher in this sample. If the observations are not paired, an independent-samples test would be appropriate instead.

Problem 13: Call Centre Response Time

Point estimate of population mean = 4.8 minutes.

95% confidence interval (large-sample t, $df = 99$): $4.8 \pm t(0.975,99) \times 1.5/\sqrt{100} = (4.502, 5.098)$ minutes.

Frequentist interpretation: if we repeatedly sampled 100 calls and built intervals in the same way, about 95% of those intervals would contain the true population mean. It is not a 95% probability statement about this fixed interval after it has been computed.

Problem 14: Defective Products

Estimated defect rate = 22/500 = 0.044 = 4.4%.

95% Wilson confidence interval for the defect proportion $\approx (0.0292, 0.0657)$, or 2.92% to 6.57%.

Frequentist interpretation: over repeated samples, 95% of intervals constructed by this method would contain the true defect proportion. The Wilson interval is preferable to a simple normal interval when proportions are relatively small.

Problem 15: Email Spam Detection

Bayes' theorem: P(spam | Offer) = P(Offer | spam)P(spam) / [P(Offer | spam)P(spam) + P(Offer | not spam)P(not spam)].

$= (0.80 \times 0.25) / [(0.80 \times 0.25)+(0.10 \times 0.75)] = 0.20/0.275 = 0.7273$.

The probability is approximately 72.73%. The keyword increases the chance that the email is spam, but it is not certain because some non-spam emails also contain 'Offer'.

Problem 16: Disease Diagnosis

P(disease | positive) = sensitivity×prevalence / [sensitivity×prevalence + (1−specificity)×(1−prevalence)].

= (0.95×0.02)/[(0.95×0.02)+(0.10×0.98)] = 0.019/0.117 = 0.1624, or 16.24%.

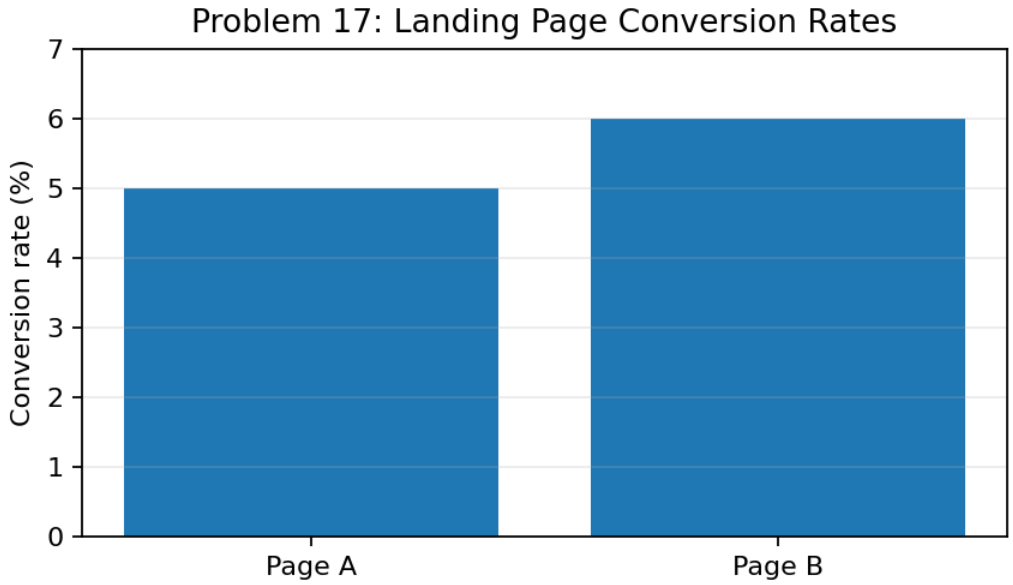
This is much lower than 95% because sensitivity is P(positive | disease), not P(disease | positive). The disease is rare, so false positives among the much larger healthy group contribute substantially.

Problem 17: Website Conversion Rate

Conversion rates: Page A = 250/5000 = 5.00%; Page B = 312/5200 = 6.00%.

Hypotheses: H_0: pA = pB; H_1: pB > pA. Pooled proportion = 562/10,200 = 0.05510. Two-proportion $z \approx 2.213$. One-sided $p \approx 0.0135$ (two-sided $p \approx 0.0269$).

At $\alpha = 0.05$, reject H_0 for the one-sided test: Page B has a statistically higher conversion rate. The absolute improvement is 1 percentage point; assess implementation cost and business value as well.



Problem 18: Mobile App Feature Test

Test H_0: $\mu B - \mu A = 0$ versus H_1: $\mu B - \mu A > 0$. Difference in means = 2 minutes.

Welch standard error = $\sqrt{(5^2/400 + 6^2/420)} = 0.385$ minutes. $t \approx 5.195$, $df \approx 803.95$. One-sided $p \approx 1.30 \times 10^{-7}$ (two-sided $\approx 2.60 \times 10^{-7}$).

Reject H_0 at $\alpha = 0.05$. Feature B shows significantly higher mean engagement in this comparison. Confirm random assignment and comparable user groups before attributing the difference to the feature.

Problem 19: Recommendation System

Old CTR = 4.8%; new CTR = 5.3%; each group has 100,000 users. H_0: pnew = pold; H_1: pnew ≠ pold.

Pooled proportion = 0.0505. Two-proportion $z \approx 5.106$; two-sided $p \approx 3.29 \times 10^{-7}$ (one-sided $\approx 1.65 \times 10^{-7}$). The difference is statistically significant.

Practical effect: CTR increased by 0.5 percentage points, about a 10.4% relative increase from 4.8%. Whether this is worthwhile depends on revenue, user experience, latency, and implementation cost.

Problem 20: Salary Prediction Model

Define paired difference as MAE(A) − MAE(B), so a positive difference favours B. Given n = 50, mean difference = 200, SD = 350.

Paired $t = 200/(350/\sqrt{50}) = 4.041$, df = 49. One-sided $p \approx 0.000094$ (two-sided $p \approx 0.000188$).

Reject H_0 at $\alpha = 0.05$. Model B's MAE is significantly lower on the paired folds. The mean reduction is 200 MAE units; check whether this improvement is practically important relative to salary scale and fold-to-fold variability.